

Fear of Temperature — Deep Research Round Two: Historical Semantic Validation, Cross-Voice Relational Modelling, and Retrieval-Specification Freeze

Executive Decision and Input Receipt

Executive decision: the historical-semantic model is sufficiently mature for a provisional implementation contract, but the research specification cannot be frozen for production collection. Seed Document A is now available and its complete 180-row Round 1 candidate inventory can be reconstructed. Seed Document B is also available and its narrative PDF exposes all 180 Priority Candidates and all 36 Expansion Candidates, giving the expected 216 Round 1B lexical records. However, the standalone Round 1B workbook, original 180-row Priority CSV, structured Expansion table, source-coverage matrix, negative register, source registry, and machine-readable JSON inventory referenced by Seed B were not supplied in this session and were not recovered from the connected Drive search. The compact PDF tables do not expose the original Round 1B candidate identifiers or all row-level fields such as relevance, confidence, expression mode, genre, ambiguity, and provenance. [filecite:turn0file0](#) [filecite:turn0file2](#)

That distinction is decisive. The two narrative inventories permit a **surface-level and anchor-level reconciliation**, but they do not satisfy the hard requirement that every Round 1B record's original identity and provenance be reconstructable. Accordingly, this report does **not** invent a complete authoritative Round 1 ↔ Round 1B crosswalk. It provides a 360-row accounting crosswalk using explicitly labelled surrogate Round 1B IDs and leaves relationships that cannot be proved from the supplied records as **UNRESOLVED**. This is consistent with the provenance principle already established in the project's data-governance work: derived or corrected states should remain auditable rather than overwrite or silently replace the evidence from which they were produced. [filecite:turn0file3](#)

Codex engineering therefore divides into two classes. Logical-schema prototyping, type definitions, database migrations in a non-authoritative development branch, and collector-interface scaffolding can safely use the contract below. **Production retrieval, pilot extraction from external corpora, query-run auditing, or large-scale collection should not begin** until the missing structured Round 1B package is restored and its original IDs are reconciled against the surrogate records created here.

Input Receipt

Input	Retrieved?	Actual recoverable records	Integrity finding
Seed A — <i>Historical Lexical Discovery and Anchor Validation</i>	Yes	180 candidates	Complete candidate IDs recoverable; six anchors × 30. SHA-256 58f89b05dd5a0a0ea49770f3dc131bd0e402fad903b25a56d03fcd8 filecite turn0file2
Seed B — <i>Stratified Lexical Expansion... Round 1B</i>	Yes	180 Priority + 36 Expansion = 216 lexical records	All forms recoverable from PDF tables/prose; original structured candidate complete per-row metadata are not exposed. SHA-256 a599da823996716c2273cb83ad52d24b3af33c9e0e943d02f5bd528 filecite turn0file0
Round 1B complete workbook	No	0	Referenced by Seed B as a downloadable structured package, but the file is absent. filecite turn0file0
Round 1B Priority Candidate CSV	No	0	Standalone 180-row CSV absent.
Round 1B structured Expansion table	No	0	Thirty-six forms recoverable only from narrative prose.
Round 1B source-coverage matrix	No standalone file	Narrative only	Coverage audit is visible in Seed B, but its machine-readable artifact is absent. filecite turn0file0
Round 1B negative/failed-evidence register	No standalone file	Narrative only	Negative evidence exists in the report but not as the referenced structured artifact. filecite turn0file0
Round 1B source registry	No standalone file	Partial reconstruction possible	Source IDs appear in compact tables; full source metadata cannot be recovered.
Round 1B machine-readable JSON inventory	No	0	Referenced by Seed B but absent. filecite turn0file0

Input	Retrieved?	Actual recoverable records	Integrity finding
Supporting conceptual/heritage review	Yes	39-page report	SHA-256 e398d7c05d06267af31da9b824586bb08a9e97e864a49fe1b4bd750 filecite turn0file1
Supporting provenance/database architecture	Yes	58-page report	SHA-256 73e611dba54b697c1188e9ce40ec8f4dbbf7779544635ff2a648f1e filecite turn0file3
Supporting feasibility assessment	Yes	34-page report	SHA-256 d8115fd3b43e3a820c2cc3d66500844962867ce4d1f2a16877a6268 filecite turn0file4

An additional integrity point matters historically. Seed B itself recorded that, **when Round 1B was originally produced**, the named Round 1 report and exact 180-row inventory were unavailable, so Round 1B deliberately refrained from fabricating a 180-row mapping and worked from 37 recoverable Round 1 terms/probes instead. The current session improves that situation because Seed A is now present, but it does not retrospectively create the missing structured Round 1B identity fields.
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The machine-readable Round 2 package generated from the material actually available contains all seventeen requested artifacts plus an output-checksum file:

[Download the Round 2 artifact bundle](#)

The bundle contains round2_input_receipt.csv, round1_round1b_crosswalk.csv, canonical_concepts.csv, lexical_forms.csv, lexical_form_senses.csv, evidence_passages.csv, semantic_relations.csv, social_relations.csv, query_rules.csv, exclusion_rules.csv, source_registry.csv, negative_unresolved_register.csv, pilot_sampling_plan.csv, annotation_schema.json, data_dictionary.md, research_freeze_manifest.json, and README.md, plus OUTPUT_CHECKSUMS.csv. Surrogate identifiers beginning R1B-P- or R1B-X- are explicitly marked as reconstruction aids and must not be mistaken for original Round 1B IDs.

Reconciliation and Semantic Method

Round 1 ↔ Round 1B accounting

Round 1 contains 180 original candidate IDs. Round 1B contains 180 ranked Priority forms and 36 Expansion forms. The generated accounting ledger therefore preserves **396 seed-stage lexical records** rather than collapsing duplicates across rounds: 180 Round 1 records + 180 Round 1B Priority records + 36 Expansion records. This is intentional. An identical string occurring in two research rounds is two provenance-bearing research records until an explicit identity relation says otherwise.
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The surface-normalised reconciliation produced:

Reconciliation result	Records
Exact Round 1 ↔ Round 1B Priority surface matches	36
Round 1 records without provable exact Priority identity	144
Round 1B Priority records without provable exact Round 1 identity	144
Round 1B Expansion records	36
Total accounting crosswalk rows	360

The 36 exact-string joins should not be confused with Seed B's statement that it had re-examined 37 Round 1 terms/probes. A “probe” in the Round 1B methodological account need not have become an exact-string Priority record; one may be a negative or differently normalised probe. The discrepancy therefore cannot be resolved responsibly without the missing structured Round 1B files.

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The current reconciliation decisions are correspondingly conservative: the 36 exact matches are **RETAIN**; all 324 other crosswalk rows remain **UNRESOLVED** at the identity level. No record disappears. In the broader lexical-form ledger, inherited negative and conditional evidence is also retained under explicit statuses rather than being discarded: **EVIDENCE_ONLY**, **QUERY_VARIANT**, **NEGATIVE_CONTROL**, **UNRESOLVED**, or **EXCLUDE**.

Analytical unit and controlled dimensions

The primary unit remains **a source-linked passage**, not the isolated word. Each accepted passage must identify or explicitly mark unknown: document/source, date, speaker, voice, expression mode, genre, lexical occurrence, temperature/climate referent, affected subject, discourse function, affect status, risk/threat status, and sufficient surrounding context. This follows the central Round 1B correction that source container and substantive speaker are separate: a scientist quoted in a newspaper remains V1, and a survey respondent represented in an academic article remains V5. filecite turn0file0

The four layers remain separate:

Layer	Operational role
A — Temperature / Physical Phenomenon	Measurement, thermal state, heat, warming, heatwaves, physical thresholds.
B — Climate / Atmospheric / Causal	Climate, atmosphere, CO ₂ , greenhouse mechanism, human attribution.
C — Affect	Explicit or plausibly affective fear, worry, anxiety, concern, distress, hopelessness, etc.
D — Threat / Risk / Harm	Danger, threat, risk, crisis, mortality, damage, vulnerability and consequences.

A Layer-D occurrence does not become Layer C merely because danger could cause fear. The supporting conceptual review makes the same distinction: perceived threat is an appraisal, whereas fear, worry and anxiety are affective or affective-cognitive states; high salience or threatening language does not demonstrate audience emotion. filecite turn0file1

The five voices remain V1 Scientific/Research, V2 Institutional/Governance, V3 Mediated Public, V4 Organised Civic/Advocacy, and V5 Direct Public/Lay. Expression mode remains E1 Direct/Spontaneous, E2 Quoted/Mediated, E3 Elicited, E4 Researcher-Coded/Labelled and E5 Not Applicable. In particular, **survey-question wording, response-option wording, participant endorsement, participant-generated wording and spontaneous public expression are separate objects.**

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Historical retrieval windows

The recommended windows are:

Anchor	Strict anchor window	Contextual validation window	Rationale
1842	1 Jan–31 Dec 1842	1839–1845	Keeps the baseline tied to 1842 while permitting attribution of neighbouring meteorological, sanitary and correspondence evidence.
1938	1 Jan–31 Dec 1938	1936–1940	Preserves Callendar's 1938 conjunction as the anchor without turning 1949 into a seventh anchor.
1988	1 Jan–31 Dec 1988	1986–1990	Captures immediate scientific, institutional and media framing around the 1988 conjunction.
2006–2007	1 Jan 2006–31 Dec 2007	2005–2008	The anchor is intrinsically a two-year bridge, combining AR4-era assessment, Stern, media and survey evidence.
2015	1 Jan–31 Dec 2015	2014–2016	Keeps Paris threshold governance strict while allowing immediate negotiating and reception context.
2022	1 Jan–31 Dec 2022	2021–2023	Keeps the contemporary endpoint bounded while allowing instrument/publication lag around affect studies.

Anything outside the contextual window must be stored as **later/earlier historical interpretation**, not silently upgraded to near-anchor evidence.

Comparability rule

STRONGLY_COMPARABLE requires a defensible concept relation, preserved source context, compatible referent/function, and explicit voice/genre differences. **PARTIALLY_COMPARABLE** means a shared semantic dimension exists but function, scale, voice, or provenance differs materially.

NOT_COMPARABLE records an invalid historical equivalence rather than a failed search. **UNRESOLVED** is used where provenance or source context is insufficient.

This rule prevents precisely the false chronological narratives rejected by both seed rounds: 1842 embodied heat cannot be retrospectively called “climate anxiety”; 1938 heatwave suffering cannot be treated as public fear of Callendar's CO₂ proposition; and 2015 “common concern” cannot be treated as the treaty parties reporting an emotion. filecite turn0file0 filecite turn0file2

Lexical Families and Semantic Case Studies

Family adjudication

Family	Historically defensible canonical distinctions	Cross-anchor decision	Retrieval consequence
F1 Temperature / Threshold	Local/instrumental temperature; mean temperature; global aggregate temperature; 2°C threshold; 1.5°C threshold.	1842↔1938 measurement can be strong when scale is explicit; measurement↔2015 governed threshold is only partial.	Never retrieve temperature without environmental/measurement context. Preserve baseline and spatial scale.
F2 Heat	Physical heat source/state; embodied exposure; heatwave/weather event; extreme-heat hazard.	Related family, not one stable sense.	heat requires environment/weather/occupation context; exclude cooking, unrelated industrial and metaphorical heat.
F3 Warming	Temperature increase; climate-system warming; global warming as issue label.	Strong/partial according to measurement versus issue-label function.	Record whether the occurrence is an observation, process, causal outcome or public issue label.
F4 Climate	Historical climatological condition; change of climate / climatic change ; modern climate-change issue; climate system; changing atmosphere.	Highest false-continuity risk.	climate alone is high-review; exclude social/business/political climate. Same spelling never proves the same concept.

Family	Historically defensible canonical distinctions	Cross-anchor decision	Retrieval consequence
F5 Carbon / Greenhouse	Atmospheric CO ₂ ; anthropogenic CO ₂ production; greenhouse mechanism; anthropogenic GHG emissions.	Mechanism, attribution and policy framing are related but not equivalent.	Require atmospheric/climate context and distinguish explanation from governance.
F6 Concern / Alarm	Public alarm; personal/ evaluative concern; institutional evaluation; common concern of humankind legal formula.	Legal formula is not affectively comparable to personal concern.	Store USED_AS_LEGAL_FORMULA separately; concern requires speaker and object.
F7 Worry	Media-prescribed worry; questionnaire worry wording; participant endorsement; spontaneous worry.	Same affect family but social functions can be non-comparable.	Split instrument and response occurrences; never infer E1 from E3.
F8 Fear / Afraid	Direct fear; participant endorsement of afraid; supplied/ research-coded fear categories; historical negative probes.	No validated linear historical continuity.	Require experiencer + climate/ heat target + wording provenance. Threat is not sufficient.
F9 Anxiety	anxious; anxiety symptom measure; climate-anxiety construct; eco-anxiety construct; self-identification where demonstrated.	Modern constructs cannot simply be projected backward.	Provenance must show researcher label, instrument category, endorsement or participant-generated wording.

Family	Historically defensible canonical distinctions	Cross-anchor decision	Retrieval consequence
F10 Distress / Depression	Historical bodily/energetic “depressing effect”; modern psychological distress; distress; depressed affect.	1842↔2022 is NOT_COMPARABLE as the same psychological construct.	Mandatory false-friend review for <code>depress*</code> and <code>distress</code> .
F11 Danger / Threat	Hazard evaluation; institutional climate threat; personal/community threat appraisal; security frame.	Partial comparison only when threatened object/group is recorded.	Require threat object and affected subject. No automatic affect label.
F12 Risk	Probabilistic risk; economic cost/risk; administrative risk assessment; loss-and-damage risk; personal perceived risk.	Usually partial, not exact.	Store domain and outcome; exclude generic financial/medical/military risk.
F13 Crisis / Emergency	Severity predicate; <code>climate crisis</code> issue label; emergency-response category; <code>climate emergency</code> frame/declaration.	Crisis ≠ emergency preparedness ≠ climate-emergency issue label.	Require explicit climate referent or enclosing governance section.
F14 Harm / Loss / Consequences	Bodily harm; mortality; damage; loss and damage; non-economic loss; vulnerability; future-generation harm.	Comparable as adverse consequence only after affected subject and domain are preserved.	Store who/what is harmed; harm is not affect.

These distinctions are historically supported by the anchor sources rather than imposed only for database convenience. Callendar's 1938 paper explicitly combines artificial CO₂ production, radiative reasoning and temperature change, making it a causal-science configuration rather than an affective

one. ¹ The official Paris text, by contrast, operationalises global average temperature through “well below 2°C” and pursuit of 1.5°C while separately addressing climate risks and impacts, demonstrating the governance-threshold sense. ²

Case A — 1842 bodily heat

The 1842 sanitary inquiry materially strengthens the baseline because substantive worker testimony survives inside an institutional inquiry. It describes workshops in terms including “most suffocating,” workers who could “faint away” under excessive heat and closeness, and a “very depressing effect on the energies.” In the surviving transcription, those expressions concern embodied occupational heat, ventilation, physiological burden and diminished capacity. They do **not** constitute evidence that workers expressed fear of climate. The transcription inspected here was produced from Internet Archive page images; it is therefore recorded as an inspected transcription derived from a facsimile source, not as direct inspection of the original physical volume. ³

The appropriate annotations are:

- **most suffocating** : V5/E3, **EXPERIENCE_REPORT** , primarily embodied heat exposure with harm-adjacent evaluation; **NO_AFFECT** or, if the full sentence warrants it, **AFFECT_ADJACENT** .
- **faint away** : V5/E3, **HARM_REPORT** , Layer D, **NO_AFFECT** .
- **depressing effect** : V5/E3, historically bodily/energetic evaluation, **AFFECT_ADJACENT** , and **FALSE_FRIEND** relative to modern psychological depression.
- **complain of the heat** : V5/E3, experience/evaluation; it shows dissatisfaction or burden but not fear.

Thus the important Round 2 result is not “1842 already had climate emotion.” It is that **lived heat language becomes visible once direct-public testimony is distinguished from the institutional container**, while modern climate-affect categories remain unvalidated. That is consistent with Seed B's correction of the earlier meteorological-only picture. [filecite turn0file0](#)

Case B — 1938 dual discourse

Callendar's scientific discourse and contemporary heat-event journalism belong in the same historical anchor but not in one causal-affective narrative. Callendar's article uses an anthropogenic-atmospheric configuration—artificial CO₂ production, atmospheric/radiative mechanism, mean-temperature change and observational comparison. The publisher record confirms the 1938 article and its scientific framing.

¹

By contrast, Round 1B's 1938 media stratum includes heat-wave, drought, scorching-wind, fire and bodily-severity language. Seed B properly treats part of this evidence as moderate where the original newspaper page was not directly inspected. [filecite turn0file0](#) The two corpora are **partially comparable at the level of physical heat/temperature**, but **NOT_COMPARABLE** as evidence that the public feared Callendar's anthropogenic-warming proposition. The date alone creates no semantic relation.

The difficult phrase **rather a coincidence** belongs to epistemic reception/evaluation. Unless an original speaker turn establishes affective content, it should be coded **NO_AFFECT** , not as worry, sceptical anxiety or fear.

Case C — 1988 public-facing versus public-generated discourse

The 1988 anchor contains a defensible multi-voice conjunction. Hansen's scientific work makes warming, greenhouse mechanisms, observations and modelling part of V1 scientific attribution; the Toronto Conference places atmospheric change in a V2 institutional register that includes security, threat, consequences and crisis; journalism translates parts of that scientific-institutional configuration into V3 public-facing headlines and warnings. Hansen's contemporaneous scientific paper is independently documented by NASA, while the Toronto Final Statement is preserved in the UN documentary record and archival reproductions. ⁴

The semantic conclusion is deliberately negative with respect to V5: **Round 1B did not locate sufficiently strong exact-1988 direct-public evidence to reconstruct lay fear, worry or anxiety at the same evidential standard.** Media intensity does not fill that empty cell. `global_security`, for example, should be `THREAT_WARNING` / `INSTITUTIONAL_EVALUATION` or `THREAT_WITHOUT_AFFECT`, not public fear.

Case D — 2006–2007 “worried”

TIME's 3 April 2006 cover package is a paradigmatic provenance case. The publication itself tells its readership, “Be Worried. Be Very Worried.” That is V3/E1 **media-generated affect prescription**: `USED_AS_MEDIA_IMPERATIVE`, `AFFECT_PRESCRIPTION`, `MEDIA_PRESCRIBED_AFFECT`, `ADDRESSED_TO` → readers. The expression was inspected on TIME's own archive/issue pages, not through a secondary quotation. ⁵

Gallup's 2007 evidence performs a different function. Its questionnaire asks respondents how much they “personally worry” about environmental problems and supplies response levels including “a great deal.” Gallup's own page documents both the question and the 2007 fieldwork. ⁶ The correct relational model is:

Gallup instrument → SUPPLIES wording/category → worry concept

V5 respondent endorsement → ELICITED_BY → Gallup item

respondent endorsement → USED_AS_PARTICIPANT_ENDORSEMENT → worry

The instrument's words are **not** participant-generated vocabulary. Consequently, TIME's imperative and a Gallup respondent's category endorsement are `RELATED_NOT_EQUIVALENT` and **not comparable as expression mode**, even though both involve the worry family.

Case E — 2015 “concern”

The Paris Agreement states that climate change is a “common concern of humankind.” In the official UNFCCC text this appears in the preambular legal-diplomatic register, not as a psychological measurement. ⁷ Its correct coding is V2/E1, `LEGAL_FORMULATION`, `INSTITUTIONAL_EVALUATION`, `USED_AS_LEGAL_FORMULA`. Treating it as personal concern would be a category error.

That sense must remain distinct from V5 survey worry/concern categories and from V4 mobilisation. Paris also separately formalises loss-and-damage response domains such as early-warning systems,

emergency preparedness, risk assessment and non-economic losses; those are procedural/governance objects, again not affect. ⁸

Case F — 2022 anxiety vocabulary

The contemporary endpoint contains dense affect vocabulary, but density makes provenance separation more important. Peer-reviewed 2022 work explicitly uses compounds such as `eco-anxiety` and `climate anxiety` as research constructs. Rehling's qualitative study, for example, analytically organises experiences gathered from semi-structured interviews; this does not imply that every participant independently called their experience "eco-anxiety." ⁹ Other 2022 research similarly operationalised climate anxiety as a research variable/construct. ¹⁰

The Yale/Meta international survey provides a different V5/E3 evidential channel. Its 2022 report covers respondents across 192 countries and territories, using supplied questions/categories for worry and perceived threat. Those are participant endorsements of instrument wording, not spontaneous lexical production. ¹¹

IPCC Working Group II's 2022 assessment supplies yet another channel: its discussion of heat exposure, mortality and mental-health challenges including anxiety and stress is V1 scientific-assessment language. It is evidence that anxiety/stress are recognised health-impact categories in the assessment; it is not V5 participant wording. ¹²

The necessary 2022 split is therefore:

`climate anxiety` / `eco-anxiety` as V1/E4 researcher constructs unless self-use is explicitly evidenced;

`symptoms of anxiety` as instrument/health-measure language where applicable;

`anxious`, `afraid`, `hopeless`, `depressed` as V5 only when the participant's endorsement or generated wording is separately documented;

`psychological distress` as a research/health construct unless source-level evidence demonstrates a participant's own wording.

Cross-Voice, Cross-Anchor and Relation Models

Cross-voice functional matrix

"Attested" below means recoverable in the seed and verified evidence; it is not a frequency statement.

Family	V1 Scientific	V2 Institutional	V3 Mediated public	V4 Civic	V5 Direct public	Functional contrast
F1 Temperature	Attested	Attested	Thin/ contextual	Thin	Thin/ contextual	Measurement → aggregation → governance threshold.
F2 Heat	Attested	Attested	Attested	Thin	Attested	Physical state → bodily exposure → weather event → health hazard.
F3 Warming	Strong	Strong	Attested	Attested later	Mainly elicited issue object	Scientific process becomes public/policy issue label without becoming affect.
F4 Climate	Strong	Strong	Strong later	Attested later	Attested later	Sense, scale and causal assumptions vary sharply by date and voice.
F5 Carbon/ greenhouse	Strong	Strong	Attested	Attested later	Thin	Mechanism/ attribution → policy/public explanation.
F6 Concern/ alarm	Thin/selective	Strong	Attested	Attested	Elicited	Legal/ institutional stance ≠ personal affect.
F7 Worry	Research/ instrument role	Thin	Strong in 2006	Thin	Strong E3 after 2006	Imperative ≠ instrument category ≠ endorsement ≠ spontaneous wording.

Family	V1 Scientific	V2 Institutional	V3 Mediated public	V4 Civic	V5 Direct public	Functional contrast
F8 Fear/ afraid	Research/ instrument role	Thin	Thin	Thin	Contemporary attestation; early probes negative	Thesis-title family requires direct source proof.
F9 Anxiety	Strong contemporary constructs	Health-assessment use	Thin	Thin	Survey/ interview evidence	Research construct ≠ symptom measure ≠ self-description.
F10 Distress/ depression	Modern research construct	Health context	Thin	Thin	1842 embodied testimony + modern public evidence	Major false-friend family.
F11 Danger/ threat	Attested	Strong	Strong	Attested	E3 attested after 2006	Threat function varies by threatened object and speaker.
F12 Risk	Strong	Strong	Thin	Attested	E3 personal appraisal	Scientific/ probabilistic, economic, administrative, legal and personal senses differ.
F13 Crisis/ emergency	Contextual	Strong	Strong	Strong modern	Thin	Severity predicate, formal response category and issue label must be split.
F14 Harm/ loss	Strong	Strong	Attested	Attested modern	Attested	Affected subjects move among bodies, livelihoods, ecosystems and generations.

These asymmetries are part of the evidence. Round 1B explicitly records V4 as not located for 1842 and 1938, V5 as not located for 1988, and contemporary V5 affect as strongly dependent on elicitation. Those cells must not be filled for visual symmetry. [fileciteopen0file0](#)

Cross-anchor comparability matrix

Proposed comparison	Status	Reason
1842 mean temperature ↔ 1938 mean temperature	STRONGLY_COMPARABLE when spatial/temporal aggregation is explicit	Both can function as measurement/aggregate variables.
1842 local temperature ↔ 2015 1.5°C / 2°C	PARTIALLY_COMPARABLE	Same physical dimension, radically different scale, baseline and governance function.
1842 occupational heat ↔ 2022 heat-related mortality	PARTIALLY_COMPARABLE	Both concern heat harm; exposure setting, scientific causal frame and affected population differ.
1842 climate ↔ modern climate change	NOT_COMPARABLE as the same sense	Shared lexical family does not establish modern causal/ problem framing.
1938 increase in mean temperature ↔ 2007 warming of the climate system	PARTIALLY/STRONGLY COMPARABLE depending passage	Both concern warming, but observational scope and assessment status differ.
1938 heatwave journalism ↔ Callendar's anthropogenic CO ₂ claim	NOT_COMPARABLE causally/ affectively	Co-temporality is not evidence of reception.
1988 institutional threat ↔ 2022 survey threat	PARTIALLY_COMPARABLE	Shared threat appraisal but different voice, scale, elicitation and affected subject.
TIME Be Worried... ↔ Gallup worry endorsement	NOT_COMPARABLE as mode	Media prescription versus elicited participant endorsement.
2015 common concern of humankind ↔ personal survey concern	NOT_COMPARABLE affectively	Legal formula versus individual evaluative/ affective response.
1842 depressing effect ↔ modern depression/ distress	NOT_COMPARABLE	Historical bodily/energetic sense versus modern psychological category.

Proposed comparison	Status	Reason
Early failed <code>anxiety</code> probes ↔ 2022 <code>climate</code> <code>anxiety</code>	UNRESOLVED / unavailable continuity	Negative historical validation cannot establish either absence or conceptual continuity.
1988 conflict risk ↔ 2015 loss-and-damage risk ↔ 2022 personal threat/risk	PARTIALLY_COMPARABLE	Common risk family; domains and social functions differ.

The preferred historical language is therefore **continuity, reconfiguration, change of referent, institutionalisation, psychologisation, lexicalisation, change of discourse function, emergence in sampled evidence, or unavailable comparison**. “Evolution” should not be used as the default causal story.

Semantic relation ontology

The implementation-ready ontology has three controlled groups.

Lexical/conceptual relations: `EXACT_VARIANT_OF`, `MORPHOLOGICAL_VARIANT_OF`, `HISTORICAL_FORM_OF`, `ORTHOGRAPHIC_VARIANT_OF`, `BROADER_THAN`, `NARROWER_THAN`, `RELATED_NOT_EQUIVALENT`, `SAME_FAMILY_DIFFERENT_SENSE`, `FALSE_FRIEND`, and `ANACHRONISTIC_MAPPING_REJECTED`.

For example, `depressing effect` in the 1842 worker passage is appropriately related to modern depression vocabulary through `SAME_FAMILY_DIFFERENT_SENSE` and, for any attempted clinical equivalence, `FALSE_FRIEND` or `ANACHRONISTIC_MAPPING_REJECTED`. Conversely, a genuine hyphenation difference such as `heat-wave / heat wave` can be represented as `ORTHOGRAPHIC_VARIANT_OF` without claiming a new concept.

Functional relations: `USED_AS_MEASUREMENT`, `USED_AS_CAUSAL_EXPLANATION`, `USED_AS_HAZARD`, `USED_AS_RISK`, `USED_AS_THREAT`, `USED_AS_CRISIS`, `USED_AS_EMERGENCY`, `USED_AS_GOVERNANCE_THRESHOLD`, `USED_AS_LEGAL_FORMULA`, `USED_AS_MEDIA_IMPERATIVE`, `USED_AS_RESEARCH_CONSTRUCT`, `USED_AS_INSTRUMENT_CATEGORY`, `USED_AS_PARTICIPANT_ENDORSEMENT`, and `USED_AS_SPONTANEOUS_EXPRESSION`.

Social/provenance relations: `EXPRESSED_BY`, `ADDRESSED_TO`, `REFERS_TO`, `AFFECTS_GROUP`, `MEDIATED_BY`, `ELICITED_BY`, `CODED_BY`, and `CALLS_FOR_RESPONSE`.

The exhaustive CSVs define, for every relation type, the valid subject entity, valid object entity, direction, valid example and invalid-use example. The model deliberately follows the principle that concepts and their labels are not identical objects. SKOS likewise separates concepts from labels and provides explicit broader, narrower and related relations, making it a useful interoperability reference without implying that SKOS identity automatically proves historical semantic identity. ¹³

Likewise, provenance relations should not be stored as informal notes. W3C PROV-O models entities, activities and agents and provides structures for derivation, quotation, revision and primary-source

relations; those capabilities directly support the project's need to distinguish original historical material from later quotation or reproduction. ¹⁴

Canonical Lexicon and Retrieval Specification

Canonicalisation decision

A provisional operational concept inventory has been created across F1–F14, but **none is marked final** `CANONICAL_CORE` in the freeze manifest, because final canonical status would imply that the missing structured Round 1B identities have already been reconciled. Instead, concepts are semantically specified and queryable where appropriate while their freeze decision remains `UNRESOLVED` pending input restoration.

This is a methodological rather than substantive failure. The following boundaries are sufficiently strong to be used in schema and annotation prototyping:

- `temperature`, `mean temperature`, `global aggregate temperature`, `1.5°C threshold` and `2°C threshold` remain separate concepts.
- `heat source/state`, `embodied heat exposure`, `heatwave event` and `extreme-heat hazard` remain separate.
- `temperature increase`, `climate-system warming` and `global warming` as issue label remain distinct.
- `historical climate`, `historical climatic change`, `modern climate change`, `climate system` and `changing atmosphere` are not merged.
- `concern`, `common concern of humankind`, `prescribed worry`, `elicited worry`, `direct fear`, `researcher-coded climate anxiety`, `psychological distress`, `institutional threat`, `risk and crisis` remain distinct.
- `loss and damage` is represented separately from generic `damage` and `non-economic losses`.

That design follows the seed material's core warning against reducing fear, concern, risk and threat to a single vocabulary and the supporting conceptual review's distinction between physical variables, climate frames, emotion, appraisal and rhetorical urgency. ^{filecite turn0file1}
^{filecite turn0file2}

Query model

The generated `query_rules.csv` contains **143 provisional anchor-specific retrieval rules**. Each rule carries: query ID, linked concept, anchor, surface form, query type, case policy, hyphenation and orthographic treatment, OCR policy, strict and contextual date windows, required/optional context terms, exclusions, passage window, source strata, expected voice, ambiguity/precision/recall risk, negative-control status, review priority, and valid/invalid-match patterns.

The key engineering rule is:

A retrieval hit is a candidate occurrence; it is never an evidence passage until source, speaker, referent, sense and context have been verified.

OCR variants are not generated speculatively. They should enter a query rule only after a recurrent OCR error has been confirmed against page images or an equivalent authoritative representation. This avoids manufacturing pseudo-historical spellings from search-engine noise.

High-risk ambiguity contract

Form	Valid domain	Binding exclusions / minimum evidence
temperature	Atmospheric/environmental measurement, global aggregate, governance threshold	Exclude body/fever, cooking and unrelated industrial temperatures. Require measured object or climate/environmental referent.
heat	Environmental/occupational heat, sun's heat, heatwave/hazard	Exclude recipes, ovens, unrelated process heat and political metaphors.
climate	Climatology, climate system/change	Exclude business, political, social and school climate unless historically under separate analysis.
warming	Temperature/climate-system warming	Exclude warming food, rooms and interpersonal metaphor.
concern	Personal concern, institutional evaluation, legal formula coded separately	Exclude business/matter concern; require speaker + object + discourse function.
alarm	Public alarm/reaction about heat/climate	Exclude clocks, alarms/devices and unrelated warning systems.
fear	Attributed experiencer fearing climate/heat/consequence	Require experiencer + target + quotation boundary; exclude unrelated fear and accusations of fearmongering.
worry	Media imperative, instrument wording, endorsement or spontaneous worry	Record producer of wording; exclude unrelated or negated worry.
anxiety	Climate-linked participant affect, symptom measure or named construct	Require construct/instrument/participant provenance and climate target.
distress	Climate/heat-related psychological or bodily distress	Exclude financial distress, distress signals and unrelated clinical context.
depression / depressed	Explicit climate-related affect, separately from historical depressing effect	Exclude economic, topographic and meteorological depression; anti-presentist sense check mandatory.
danger	Climate/CO ₂ /heat danger	Require danger object and threatened subject.

Form	Valid domain	Binding exclusions / minimum evidence
<code>threat</code>	Explicit climate/heat threat	Exclude military, cyber and unrelated political threat.
<code>risk</code>	Climate/heat risk, specified outcome/domain	Exclude credit, investment, surgery and unrelated insurance risk.
<code>crisis</code>	<code>climate crisis</code> or explicitly climate-linked crisis	Exclude generic economic/political/ health crisis.
<code>emergency</code>	<code>climate emergency</code> or climate-context emergency preparedness	Exclude unrelated medical/ emergency-services usage.
<code>change</code>	Only constrained constructions such as <code>climate change</code> , <code>climatic change</code> , historically reviewed <code>change of climate</code>	Standalone <code>change</code> is prohibited as a production query.

The downloadable `exclusion_rules.csv` supplies at least three valid and three invalid examples for every one of these high-risk forms, together with the minimum context required for acceptance.

Negative controls

Negative controls remain first-class records. Round 1's failed fear, anxiety, alarm, danger and risk probes at early anchors are not claims of universal absence; they mean only that the reviewed source set did not validate the required temperature/climate-linked sense at the required evidential standard. Seed A is explicit about this distinction. `filecite turn0file2`

Accordingly, a failed early `fear` query should be preserved with its date, source searched, query version and rejection reason. That evidence is valuable for evaluating retrieval precision and presentist false positives even though it is not positive affect evidence.

Context, Data Contract and Annotation Manual

Passage-context contract

Every retained passage must preserve, where applicable:

Universal metadata: title, source/publication, date and date certainty, stable source identifier or archive locator, page/section, retrieval/capture provenance, matching expression and exact passage locator.

Textual context: matching sentence plus preceding and following sentence and the enclosing paragraph. The collector may capture more text internally, but this minimum must remain available to reviewers.

Speaker context: substantive speaker, quotation boundary, attribution phrase and the documentary producer. A newspaper quotation therefore stores both the newspaper and the quoted scientist/ resident without converting the quoted speaker into V3.

Genre overrides:

- A treaty phrase stores the enclosing article/paragraph; preamble or definitions are included where they control legal meaning.
- A survey occurrence stores the exact question, response alternatives, question order/construct context where material, and whether the evidence is an individual response, aggregate response, question text or researcher variable.
- A newspaper quotation stores headline/deck where analytically relevant, quotation boundaries and attribution before/after the quotation.
- A diary or letter stores the surrounding paragraph and an adjacent paragraph if pronoun/referent resolution requires it.
- Testimony stores the speaker turn and question-answer boundary.
- A scientific measurement/causal claim stores section/table/figure-caption context when the meaning depends on it.
- An OCR-derived match stores a page-image locator or equivalent visual verification reference.

This is why `Be Worried. Be Very Worried.` and `personally worry` cannot enter one undifferentiated “worry hit” table: the first is a publication-generated imperative, while the second is part of a survey instrument whose participant response must be represented separately. ¹⁵

Deduplication and source hierarchy

Duplicate handling is relational, not destructive.

A persistent identifier is checked first. Where none exists, document-level fingerprints/content hashes are combined with title, date, source and issue metadata. Exact and near-identical newspaper reprints are retained as separate manifestations inside a syndication/reuse cluster. The earliest or original manifestation is designated canonical only where provenance proves it.

Repeated quotations likewise remain separate events: the historical original, an archival transcription and a later newspaper or scholarly reproduction may contain the same wording but are not the same source. `MEDIATED_BY`, quotation/derivation relations and source-chain provenance preserve the connection. A later reproduction can support discovery or interpretation but must never be silently upgraded to original primary evidence. PROV-O's explicit distinction among primary-source, quotation, derivation and revision relations provides a suitable interoperability pattern. ¹⁴

Repeated treaty language and survey wording are represented as version/reuse relations, not deleted as duplicates. This is analytically necessary because the repetition itself may establish institutional continuity while individual administrations, translations or survey waves remain distinct evidence objects.

PostgreSQL-ready logical contract

The logical model contains the following core entities:

`historical_anchor`, `lexical_family`, `canonical_concept`, `lexical_form`,
`lexical_form_sense`, `voice`, `expression_mode`, `source_genre`, `source`, `document`,
`speaker_actor`, `audience`, `affected_group`, `evidence_passage`, `lexical_occurrence`,
`semantic_annotation`, `semantic_relation`, `social_provenance_relation`, `query_rule`,
`query_run`, `review_decision`, `negative_control`, `provenance_event`, and
`research_version`.

The essential cardinalities are:

`canonical_concept` $\leftarrow$ M:N $\rightarrow$ `lexical_form` through `lexical_form_sense`;

`document` $\rightarrow$ 1:N `evidence_passage`;

`evidence_passage` $\rightarrow$ 1:N `lexical_occurrence`;

`lexical_occurrence/evidence_passage` $\rightarrow$ 1:N `semantic_annotation`;

relations $\rightarrow$ exactly identified supporting evidence;

query rule $\rightarrow$ many versioned query runs;

evidence/occurrence $\rightarrow$ multiple immutable review decisions through supersession.

This directly enforces the project principles. One spelling can have several historical senses; one concept can have several lexical forms; one passage can contain several occurrences; voice and genre occupy different columns/tables; researcher, instrument and participant language are distinguishable; negative and rejected records remain queryable; later interpretation does not overwrite historical evidence; and every extraction/review state is versioned. These principles are also consistent with the project's existing provenance-aware database architecture, which recommends PostgreSQL at the relational core, explicit provenance events and immutable review history rather than a graph database as the primary store.

filecite turn0file3

JSONB is appropriate for irregular source-specific metadata or collector response payloads, but **not** for controlled dimensions whose integrity should be enforced relationally—voice, mode, anchor, layer, review state and relation type should use foreign keys or constrained vocabulary identifiers. PostgreSQL's JSON support makes mixed structured/unstructured metadata feasible without abandoning relational constraints. ¹⁶

The supplied `annotation_schema.json` uses JSON Schema 2020-12 conventions so that collector and annotation payloads can be validated independently of the SQL implementation. ¹⁷

Annotation manual

The required human sequence is:

verify source/date $\rightarrow$ verify lexical match $\rightarrow$ capture context $\rightarrow$ identify speaker $\rightarrow$ assign voice $\rightarrow$ assign expression mode $\rightarrow$ identify genre $\rightarrow$ identify referent $\rightarrow$ establish directness $\rightarrow$ assign provisional sense $\rightarrow$ distinguish affect from threat/risk $\rightarrow$ identify audience/affected group where explicit $\rightarrow$ assign relations $\rightarrow$ check false positives $\rightarrow$ decide retain/reject/uncertain $\rightarrow$ record confidence and reason.

The controlled review outcomes are exactly:

`ACCEPT`, `ACCEPT_WITH_QUALIFICATION`, `REJECT_FALSE_POSITIVE`, `REJECT_WRONG_SENSE`, `REJECT_WRONG_DATE`, `REJECT_WRONG_VOICE`, `REJECT_INSUFFICIENT_CONTEXT`, `DUPLICATE`, `UNRESOLVED`, and `ESCALATE_FOR_ADJUDICATION`.

Worked difficult examples:

Example	Coding decision
1842 depressing effect	ACCEPT_WITH_QUALIFICATION ; V5/E3; historical bodily/energetic sense; AFFECT_ADJACENT ; FALSE_FRIEND with clinical depression. ¹⁸
1938 rather a coincidence	ACCEPT_WITH_QUALIFICATION ; scientific epistemic evaluation, normally NO_AFFECT ; original discussion context should be checked before stronger classification. ¹⁹
1988 global security	ACCEPT ; V2; security/threat frame; THREAT_WITHOUT_AFFECT . ¹⁹
2006 Be Worried. Be Very Worried.	ACCEPT ; V3/E1; AFFECT_PRESCRIPTION ; MEDIA_PRESCRIBED_AFFECT . ²⁰
2007 personally worry	Split the instrument occurrence from respondent endorsement; V5 respondent evidence is E3, not spontaneous vocabulary. ²¹
2015 common concern of humankind	ACCEPT ; V2/E1; LEGAL_FORMULATION ; not personal emotion. ²²
2022 climate anxiety	ACCEPT as V1/E4 research construct unless the underlying passage independently proves participant self-use. ¹⁰
2022 very worried	V5/E3 participant endorsement when tied to the survey response; wording remains instrument-supplied. ²³

Pilot Plan, Acceptance Gates and Limitations

Coverage-aware approximately 200-passage pilot

The pilot target is **200 accepted source-linked passage records**, not 200 raw keyword hits. “Accepted” here means that the passage is valid research evidence; approximately twenty may deliberately be accepted **negative-control evidence** showing a wrong sense, failed historical probe or exclusion pattern. Such records do not count as positive lexical attestations.

Recommended anchor allocation:

Anchor	Accepted passage target	Voice allocation	Rationale
1842	30	V1 12; V2 3; V3 3; V4 0; V5 12	Preserve meteorological plus embodied worker/public evidence; no invented organised-civic stratum.
1938	28	V1 16; V2 2; V3 10; V4 0; V5 0	Causal science plus event journalism; no fabricated lay reception of Callendar.

Anchor	Accepted passage target	Voice allocation	Rationale
1988	30	V1 12; V2 12; V3 6; V4 0; V5 0	Strong science/governance/media conjunction; preserve V5 gap.
2006–2007	36	V1 7; V2 7; V3 6; V4 4; V5 12	Best anchor for cross-voice comparison and E1/E3 worry distinction.
2015	38	V1 4; V2 14; V3 5; V4 6; V5 9	Governance threshold/legal terminology plus civic/survey evidence.
2022	38	V1 10; V2 6; V3 4; V4 4; V5 14	Dense affect/risk endpoint while limiting contemporary dominance.
Total	200	V1 61; V2 44; V3 34; V4 14; V5 47	Coverage follows recoverability, not artificial equality.

The absence of V4 in the two early anchors and V5 in the 1988 allocation follows the Round 1B coverage audit rather than a preference for unbalanced sampling. Seed B explicitly finds those strata not located or too thin for confident reconstruction. [filecite:turn0file0](#)

Family targets are deliberately **non-additive**, because a single passage can validly contain, for example, F3 warming, F5 CO₂ causality and F11 threat. Suggested minimum exposure is F1 18; F2 18; F3 16; F4 18; F5 14; F6 12; F7 14; F8 12; F9 16; F10 12; F11 18; F12 18; F13 12; F14 18.

Source-genre targets sum to 200: scientific article/assessment 45; institutional/legal/governance 42; media 34; survey instrument/report 32; testimony/letter/diary/direct-public 31; civic/advocacy 16. These are planning targets, not quotas that override historical evidence.

Review priority should be approximately 90 high-risk, 80 medium-risk and 30 low-risk passages. E1, E2, E3 and E4 must each appear where historically recoverable. E5 should be used only where the production-mode distinction genuinely does not apply.

Pilot diagnostics

The pilot should calculate:

accepted-match precision; false-positive rate; wrong-sense rate; wrong-voice rate; insufficient-context rate; duplicate rate; provenance-completeness rate; unresolved rate; reviewer agreement; and family-specific ambiguity rate.

At least **50 of the 200 passages** should be independently double-coded, stratified across all six anchors, all fourteen families, every recoverable expression mode and each high-risk ambiguity class. This is preferable to double-coding only “easy” random records because the pilot's central purpose is to test the semantic boundary conditions that could make later automated collection misleading.

The following are **project-specific engineering gates, not universal methodological laws**:

Gate	Recommended decision rule	Why
Provenance completeness	100% for accepted positive passages on source/document/date/locator and evidence basis for voice/mode	Missing provenance defeats the thesis's principal methodological safeguard.
Overall positive retrieval precision	Target ≥90% after rule tuning	Makes manual review feasible while still allowing exploratory recall.
High-risk-family precision	Target ≥80% per high-risk family	<code>climate</code> , <code>fear</code> , <code>risk</code> , <code>concern</code> , <code>depress*</code> and similar forms are intrinsically polysemous.
Unresolved rate	≤5% overall , with no unresolved systemic boundary in fear/worry/anxiety versus threat/risk	High unresolvedness would mean the ontology or passage windows are not operational.
Wrong-voice rate	≤5% after adjudication	Voice confusion would invalidate the cross-voice research question.
Duplicate contamination	≤3% after deduplication pass	Prevents syndicated/repeated text from masquerading as independent evidence.
Double-coded raw agreement	Aim for ≥85% on core categorical decisions	A transparent project trigger for manual revision.
Agreement statistic	Cohen's κ around 0.70 or better on voice/mode/affect status as a warning threshold, not a universal validity criterion	Low agreement should trigger manual refinement rather than automatic acceptance.

A query rule should be revised when the pilot reveals a **recurrent false-positive mechanism**, not merely one idiosyncratic failure. Two or more independent instances of the same wrong-sense pattern are sufficient to open a rule review; whether the exclusion becomes automatic depends on its effect on plausible recall.

A canonical concept should change only when source-level evidence shows that its operational boundary is wrong—for example, repeated independent evidence that one surface form consistently divides into two senses the current model has merged. A single surprising hit should normally trigger `ESCALATE_FOR_ADJUDICATION`, not ontology redesign.

Conditions that block large-scale collection

Large-scale collection must remain blocked when any of the following holds:

the original seed-record identity map is unresolved;

accepted positive records lack mandatory provenance;

voice versus source genre cannot be consistently distinguished;

instrument versus participant wording remains conflated;

high-risk lexical families fail pilot precision after reasonable context rules;

OCR variants are being admitted without image-level validation;

duplicate/reprint handling cannot recover source lineage;

or reviewers cannot consistently distinguish affect, threat, risk, legal stance and researcher constructs.

The first of those conditions is present now. This alone prevents a final PASS.

Research limitations

This design deliberately preserves source asymmetry. The six anchors are not six representative samples of whole societies. They are strategically selected historical positions with very different documentary infrastructures. Seed B's coverage audit shows, for example, strong direct-worker testimony in 1842, unusually strong V1 causal science in 1938, strong V1/V2/V3 but poor V5 reconstruction in 1988, strong elicited-public evidence by 2006–2007, governance-heavy evidence in 2015, and much denser but heavily instrumented contemporary affect evidence in 2022.

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Nor does lexical availability measure prevalence. No Google Books Ngram analysis, word-frequency comparison, percentage-change calculation, co-occurrence count, embedding analysis, topic modelling, sentiment scoring, social-network analysis, large-scale scraping or causal inference was performed in this round.

The thesis proposition therefore remains investigable rather than pre-confirmed. The evidence supports a history in which **measurement, experienced heat, causal explanation, institutional danger, media warning, civic mobilisation, elicited public appraisal and named affect become differently related in different documentary systems**. It does not yet support the simpler statement that societies moved uniformly from “no fear” to “fear.” That conclusion is consistent with both seed rounds and with the broader feasibility work, which rejects a seamless global fear series. filecite turn0file2
filecite turn0file4

References and Research-Specification Freeze Manifest

Primary historical sources

Chadwick, E. (1842). *Report on an Inquiry into the Sanitary Condition of the Labouring Population of Great Britain*. The wording used in this report was inspected through a Project Gutenberg transcription produced from Internet Archive page images; it was **not** treated as direct inspection of the original physical volume. The transcription preserves the workshop testimony underlying **most suffocating**, **faint away** and **depressing effect**. ³

Callendar, G. S. (1938). “The artificial production of carbon dioxide and its influence on temperature.” *Quarterly Journal of the Royal Meteorological Society*, 64, 223–240. Publisher metadata and primary article material were inspected; the article supports the anthropogenic-CO₂/mean-temperature causal bridge rather than an affective claim. ¹

Hansen, J., et al. (1988). Contemporary scientific work on global climate change and the greenhouse effect. NASA's archival scientific record was used for the scientific framing; media reproductions should not substitute for an original testimony transcript when exact quoted wording is required. ²⁴

Toronto Conference on the Changing Atmosphere (1988). Final statement/documentary record. The UN bibliographic record and archival reproductions were inspected for institutional framing; reproduced extracts remain identified as reproductions rather than silently labelled original manuscript evidence. ²⁵

United Nations Framework Convention on Climate Change. (2015). *Paris Agreement*. The official UNFCCC PDF was directly inspected for the global-average-temperature threshold, **common concern of humankind**, early-warning, emergency-preparedness, risk and loss-and-damage formulations. ²⁶

Intergovernmental Panel on Climate Change. (2022). *Climate Change 2022: Impacts, Adaptation and Vulnerability*, Working Group II Summary for Policymakers. Official IPCC text was inspected for heat exposure, heat-related mortality and mental-health categories including anxiety and stress. ¹²

Institutional and participant/public sources

Gallup. (2007). U.S. environmental worry survey material. Gallup's own survey page was inspected for question wording, field dates and response categories; participant endorsement is coded separately from the questionnaire's supplied wording. ⁶

Leiserowitz, A., et al. / Yale Program on Climate Change Communication and Meta. (2022). *International Public Opinion on Climate Change*. The official report and methodology were inspected for worry, threat and sampling/provenance. ¹¹

Media/public sources

TIME. (2006, April 3). "Global Warming: Be Worried. Be Very Worried." TIME's own archival issue/cover pages were inspected. The phrase is therefore classified as publication-generated V3 affect prescription rather than public-generated affect. ⁵

Civic/advocacy sources

Round 1B contains V4 campaign and COP-era material and explicitly notes uneven coverage across anchors; those records remain in the seed inventory and source registry reconstruction, but the missing structured Round 1B source registry prevents this report from certifying a complete V4 bibliographic ledger. ^{filecite turn0file0}

Scholarly secondary sources

Rehling, J. (2022). Research on eco-anxiety based on qualitative interviews. It is used here specifically to demonstrate the difference between participant experience and researcher-level conceptual labelling. ⁹

Wake, S. (2022). "Understanding eco-anxiety." *Nature Climate Change*. Used as evidence of contemporary research-construct formation rather than spontaneous public vocabulary. ²⁷

Contemporary cross-national climate-anxiety research likewise operationalises a research construct rather than establishing that all participants independently generate the compound expression. ²⁸

The project's existing conceptual review also draws on history-of-emotions, risk, climate-communication and heritage scholarship to distinguish fear, worry, anxiety, concern, threat appraisal, salience and heritage-making. Those distinctions are inherited here as methodological constraints rather than collapsed into a single negative-affect category. ¹³

Methodological and data-modelling sources

W3C. (2009). *SKOS Simple Knowledge Organization System Reference*. Used as an interoperability model for stable concepts, labels and explicit semantic relations; not as evidence that lexical equivalence implies historical equivalence. ¹³

W3C. (2013). *PROV-O: The PROV Ontology*. Used to ground provenance modelling of entities, activities, agents, derivations, quotations, revisions and primary-source relations. ¹⁴

PostgreSQL documentation. Current JSON/JSONB capabilities support source-specific metadata inside a relational architecture; controlled semantic fields remain relationally constrained. ¹⁶

JSON Schema. Draft 2020-12 is used for the generated annotation interchange/validation contract. ¹⁷

The project-specific database architecture supplies the broader design commitment to a PostgreSQL relational core, explicit provenance, immutable review history and standards-aware export rather than premature migration to a graph database. ¹⁵

Research-Specification Freeze Manifest

Component	Version / state
Input audit	round2-2026-08-25-partial-v0.9
Round 1 inventory	180/180 accessible and reconstructed
Round 1B narrative lexical inventory	216/216 surface records accessible: 180 Priority + 36 Expansion
Round 1B original structured identity/provenance package	MISSING / NOT CERTIFIED
Canonical lexicon	canonical-concepts-v0.9-PROVISIONAL
Semantic/social ontology	semantic-social-ontology-v0.9
Query specification	query-spec-v0.9-PROVISIONAL
Annotation schema/manual	annotation-schema-v0.9
PostgreSQL logical contract	postgres-logical-contract-v0.9
Accounting crosswalk	360 rows; 36 exact surface matches; remaining identity relations explicitly unresolved
Seed-stage lexical ledger	396 records preserved; no silent deletion

Component	Version / state
Query rules	143 provisional rules
Source-level evidence examples	20 adjudicated high-priority passages in artifact package
Pilot	200-passage, coverage-aware design ready
Production Codex collection	NOT AUTHORISED
Schema/scaffolding prototyping	Permissible only as provisional, non-authoritative engineering
Internal hard-gate status	BLOCKED_FOR_INPUT_RECONCILIATION

The unresolved dependencies are exact and finite: the original Round 1B workbook, 180-row Priority CSV, structured Expansion table, source-coverage matrix, negative/failed-evidence register, source registry and machine-readable JSON inventory must be recovered; their original candidate IDs and row-level provenance must then be mapped to the surrogate Round 1B records. Until that is done, declaring the canonical lexicon and query specification immutable would falsely certify provenance that has not been reconstructed.

The substantive methodological work nevertheless yields a clear final pre-engineering principle: **the database must represent relationships among measurement, heat experience, causal explanation, threat, governance, mediation, elicitation, mobilisation and affect rather than treating them as stages of a single transhistorical “fear” variable.** The same word can change function, different words can instantiate related concepts, and identical affect vocabulary can do different social work depending on who produced it and how it entered the documentary record. The required architecture is therefore passage-centred, provenance-aware, voice-stratified, sense-explicit and reviewable.

RESEARCH_SPEC_FREEZE=BLOCKED

¹ <https://rmets.onlinelibrary.wiley.com/doi/pdf/10.1002/qj.49706427503>

<https://rmets.onlinelibrary.wiley.com/doi/pdf/10.1002/qj.49706427503>

² ⁷ ⁸ ²² ²⁶ https://unfccc.int/sites/default/files/resource/parisagreement_publication.pdf

https://unfccc.int/sites/default/files/resource/parisagreement_publication.pdf

³ ¹⁸ <https://www.gutenberg.org/cache/epub/65090/pg65090-images.html>

<https://www.gutenberg.org/cache/epub/65090/pg65090-images.html>

⁴ ²⁴ <https://www.giss.nasa.gov/pubs/abs/ha02700w.html>

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⁵ ¹⁵ ²⁰ <https://content.time.com/time/covers/0%2C16641%2C20060403%2C00.html>

<https://content.time.com/time/covers/0%2C16641%2C20060403%2C00.html>

⁶ ²¹ <https://news.gallup.com/poll/26932/did-hollywoods-glare-heat-public-concern-about-global-warming.aspx>

<https://news.gallup.com/poll/26932/did-hollywoods-glare-heat-public-concern-about-global-warming.aspx>

⁹ <https://journals.sagepub.com/doi/10.1177/00812463221130898>

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¹⁰ ²⁷ <https://www.nature.com/articles/s41558-022-01497-0>

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- 12 <https://www.ipcc.ch/report/ar6/wg2/chapter/summary-for-policymakers/>
<https://www.ipcc.ch/report/ar6/wg2/chapter/summary-for-policymakers/>
- 13 SKOS Simple Knowledge Organization System Reference
https://www.w3.org/TR/skos-reference/?utm_source=chatgpt.com
- 14 PROV-O: The PROV Ontology
<https://www.w3.org/TR/prov-o/>
- 16 PostgreSQL: Documentation: 18: 8.14. JSON Types
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- 17 JSON Schema
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- 28 <https://doi.org/10.1016/j.jenvp.2022.101887>
<https://doi.org/10.1016/j.jenvp.2022.101887>